

Molar fraction of
Naphthalene

Mass of
Naphthalene

Mass of
Biphenyl

0.16

0.137

0.863

0.20

0.25

0.33

0.40

0.50

0.67

0.83

for $X_N = 0.16$

$$X_N = \frac{n_N}{n_N + n_P}$$

$$0.16 = \frac{x}{128.17}$$

$$\left(\frac{x}{128.17} \right) + \left(\frac{1-x}{154.21} \right)$$

$$x = 0.137$$

$$\therefore 1-x = 1 - 0.137$$

$$= 0.863 \text{ g}$$

X_N = molar fraction of
Naphthalene

n_N = moles of naphthalene

n_P = moles of biphenyl

x = mass of naphthalene

$(1-x)$ = mass of biphenyl.

Calculate other m_N for each molar fraction of
Fill the others naphthalene as the example given
above.